Software Engineering - Project Specific Requirements Specification Submission

Project Title: Task Manager Application

Description: A task management tool to organize and track work progress

Tech Stack: JavaScript (React.js + Node.js) / Python (Django)

1. Introduction

1.1 Purpose

The **Task Management System** is intended to provide users with a structured way to create, organize, monitor, and manage their personal tasks. The system is designed to support essential task-management activities such as creating tasks, assigning due dates, priorities, and categories, viewing and completing tasks, filtering and sorting them, and receiving reminders when deadlines approach. It also provides a dashboard for viewing task statistics and graphical progress, while maintaining user-specific task data across sessions.

This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Task Management System. It is intended to provide a common understanding of the system among users, developers, testers, and other stakeholders involved in the development and evaluation of the project. The document establishes the expected system behaviour, major capabilities, quality requirements, security expectations, data-handling requirements, and performance considerations.

The SRS also serves as a reference for determining whether the developed system satisfies the requirements defined for user access, personal task management, progress tracking, reminders, dashboard monitoring, data protection, and reliable data storage. The requirements include controls such as password protection, authenticated access, prevention of cross-user data access, backup and recovery, scalability, and recording timestamps for important task actions.

1.2 Scope

The scope of the Task Management System covers the management of user accounts and personal tasks throughout their normal use of the system. Users can create accounts using email and password, log in and log out, view and edit their profile information, change their passwords, and recover forgotten passwords through email verification. The system also applies email-format validation and password-strength requirements during account management.

For task management, the system supports creating tasks with mandatory titles and optional descriptions, assigning due dates, priorities, and categories, viewing task details, marking tasks as completed, and permanently deleting tasks. Existing task priorities can also be updated. Users can create, rename, and delete categories and can organize their tasks through filtering and sorting based on attributes such as status, priority, category, due date, and priority.

The system also includes due-date reminders and a dashboard that presents summary statistics and graphical representations of task progress. User and task information is persistently stored so that it remains available across sessions. Access to task information is restricted to the respective user, and inactive sessions are automatically terminated after a defined period of inactivity.

In addition to these functional capabilities, the system is expected to satisfy requirements related to performance, security, data integrity, data protection, backup and recovery, scalability, and traceability. The dashboard is expected to load within three seconds under normal conditions. Passwords are required to be stored using one-way hashing rather than as plaintext, while task-related features require authenticated access. The system must prevent users from viewing or modifying another user's data and must preserve submitted task operations without data loss.

The system further includes provisions for manual database backup and restoration, support for at least 1,000 tasks per user without exceeding the specified dashboard performance threshold, and timestamp recording for key actions such as task creation, editing, completion, and deletion. These requirements define the quality and reliability expectations within the scope of the Task Management System.

1.3 Definitions, Acronyms, and Abbreviations

Term **Definition**

SRS Software Requirements Specification; the document that defines the functional and non-functional requirements of the Task Management System.

FR Functional Requirement; a requirement describing a function or behaviour that the system is expected to provide.

NFR Non-Functional Requirement; a requirement describing a quality, constraint, or performance expectation of the system.

Task An individual item of work managed by a user, which may contain a title, description, due date, priority, category, and completion status.

User A registered person who accesses the Task Management System to manage their own tasks and profile.

Dashboard The part of the system that presents summary statistics and graphical information about a user's tasks and progress.

Authentication The process of verifying a user's identity through valid credentials before providing access to protected features.

Authorization The control that ensures a user can access and modify only the tasks and data belonging to that user.

Session The period during which a user remains logged in to the Task Management System.
Priority A level assigned to a task to indicate its relative importance.

Category A user-managed grouping used to organize related tasks.

1.4 Reference Documents

The following materials were used as references for preparing this SRS:

1. **Task Management System - Functional and Non-Functional Requirements Document,** the primary source containing the functional requirements FR-01 to FR-28 and non-functional requirements NFR-01 to NFR-09.
2. **SRS Template provided for the academic project** - used as the structural reference for organizing the SRS sections, including Introduction, Overall Description, Specific Requirements, Appendices, and Index.

No additional formal reference documents, standards, books, journal papers, or external sources were provided for preparing this section.

1.5 Overview

The complete SRS is organized into several sections to present the requirements of the Task Management System in a structured manner.

Section 1 - Introduction provides the purpose, scope, terminology, reference documents, and overall organization of the SRS.

Section 2 - Overall Description is intended to provide a broader description of the system, including its product perspective, major functions, user characteristics, constraints, and assumptions and dependencies.

Section 3 - Specific Requirements is intended to contain the detailed requirements of the system and forms the main body of the SRS, where the functional and non-functional requirements are specified in greater detail.

The **Appendices** section is intended to contain supporting information that may be required to complement the main requirements, while the **Index** provides an organized reference to important terms and topics within the complete SRS.

2. Overall Description

2.1 Product Perspective

The Task Management System is designed as a simple and organized application that helps users manage their personal tasks from a single place. Instead of keeping track of tasks manually or across different platforms, users can use the system to create, organize, monitor, and complete their tasks.

A user can first create an account and securely log in to the system. After logging in, the user can manage their profile and create tasks according to their needs. Each task can contain a title, description, due date, priority, category, and completion status. This allows users to keep both basic and detailed information about the work they need to complete.

The system also provides features to make task management easier. Users can filter and sort their tasks based on information such as status, priority, category, or due date. They can mark completed tasks, update task priorities, and permanently delete tasks when they are no longer required. Users can also create and manage their own categories to organize related tasks.

A dashboard gives users a quick view of their overall task progress. It displays summary statistics and graphical information, allowing users to understand their current task status without having to go through every task individually. The system also provides reminders

when a task's due date is approaching or has arrived.

The system is designed around individual user accounts. Therefore, the tasks and information belonging to one user must remain separate from those of other users. User and task information is stored persistently so that it remains available even after the user logs out and returns to the system later.

Security and reliability are also important parts of the system. Task-related features are available only to authenticated users, passwords are stored using one-way hashing, and users are prevented from accessing another user's information. The system also maintains timestamps for important task activities and supports database backup and recovery.

2.2 Product Functions

The Task Management System provides the following major functions:

User Account and Profile Management:

The system allows users to create an account using their email address and password. During registration, the email format is checked and the password must satisfy the required strength level. Registered users can log in using their credentials and log out when they finish using the system.
Once logged in, users can view and edit their profile information. They can also change their existing password by verifying the current password. If a user forgets their password, the system provides a password-reset option through email verification.

Task Creation and Management:

Users can create new tasks by providing a mandatory title and, when required, an optional description. They can assign a due date, priority, and category to each task. Users can view the complete details of their tasks, change their priority, mark tasks as completed, and delete tasks that are no longer needed.

Task Organization:

To help users keep their tasks organized, the system allows them to create, rename, and delete categories. Tasks can then be assigned to appropriate categories. Users can also filter tasks based on their status, priority, or category and sort them using information such as due date or priority.

Reminders and Progress Tracking:

The system provides reminders when a task's due date is approaching or has arrived. This helps users keep track of deadlines and reduces the possibility of overlooking important tasks.The dashboard provides a summary of the user's tasks and displays their progress using statistics and graphical representations.

Data Storage and User-Specific Access:

The system permanently stores user and task information so that the data is available across sessions. Each user's tasks are associated with their account, ensuring that a user can access only their own task information. The system also records timestamps for important actions such as creating, editing, completing, and deleting tasks.

2.3 User Characteristics

The main users of the Task Management System are individual users who want to manage their personal tasks and keep track of their work.

The system does not require users to have specialized technical knowledge. A user should have basic familiarity with computers or similar devices and should be able to perform common activities such as creating an account, logging in, entering information into forms, and navigating through an application.

Users are expected to provide their own account information and manage their tasks through the system. They should also be able to understand basic task-related information such as due dates, priorities, categories, and completion status.

Since the system is intended for personal task management, each user is responsible for maintaining and updating their own tasks and profile information.

2.4 Constraints

The development and operation of the Task Management System are subject to several constraints.

First, task-related features must be available only to authenticated users. The system must ensure that one user cannot view, change, or delete another user's tasks or personal information.

Security is another important constraint. Passwords must be stored using a one-way hashing algorithm and must never be stored as plain text. The system must also provide appropriate session handling, including automatically logging out inactive users after the defined period.

The system must provide acceptable performance during normal use. In particular, the dashboard should load within 3 seconds under normal conditions. It should also be capable of supporting at least 1,000 tasks for a single user without causing performance to exceed the specified limit.

Data integrity is also a constraint. Task operations submitted by users should not be lost,

and existing task information should not be changed unless the authorized user explicitly performs an action. The database must additionally support manual backup and successful restoration of data.

2.5 Assumptions and Dependencies

The system is developed based on a few reasonable assumptions about how it will be used.

It is assumed that users have access to a suitable device and the necessary environment to use the application. Users are also expected to provide valid information while registering and managing their accounts. A valid email address is particularly important because email verification is used for password recovery.

It is also assumed that users will manage only their own tasks and will perform actions such as creating, editing, completing, and deleting tasks through their authorized account.

The system depends on a database for storing user accounts, tasks, categories, and other related information. Persistent storage is necessary so that the user's information remains available between sessions.

The system also depends on an email service for features such as password recovery and email verification. A notification mechanism is required to provide reminders for approaching or arrived task deadlines.

Authentication and authorization mechanisms are required to ensure that protected features are accessible only to logged-in users and that users cannot access another user's information. Finally, the database must support backup and recovery so that stored information can be restored when required. These dependencies support the system's security, reliability, and normal operation.

3. Specific Requirements

External Interface Requirements

A.1 User Interfaces

The Task Management System shall provide a graphical user interface through which users perform all account and task management activities described in this SRS. The user interface shall include, at minimum, the following interaction screens/areas:

- Registration and login screens, including fields for email and password entry.
- A profile screen allowing users to view and edit their profile information.

- Password management screens for changing the current password and for initiating/completing a password reset via email verification.
- Task management screens for creating, viewing, editing, completing, and deleting tasks, including fields for title, description, due date, priority, and category.
- A category management screen for creating, renaming, and deleting task categories. • Filtering and sorting controls for organizing the task list by status, priority, category, and due date. • A dashboard screen presenting summary statistics and graphical representations of task progress.

The Task Management System shall be delivered as an Android mobile application, and all screens listed above shall be designed for a mobile device form factor and touch-based interaction.

A.2 Software Interfaces

- Email Service Interface: The system shall interface with an email service to send password-reset/verification messages, as stated in the Scope ("recover forgotten passwords through email verification"). • Database/Storage Interface: The system shall interface with a persistent data store to retain user and task information across sessions.
- Push Notification Service Interface: The system shall interface with a push notification service to deliver task due-date reminders to the user's Android device.

A.3 Hardware Interfaces

The system shall operate on Android mobile devices and shall support interaction through the device's touchscreen. No specialized hardware is required for the operation of the application.

Functional Requirements

Functional requirements are grouped below by functional area for readability. Original requirement identifiers (FR 01 to FR-28) are preserved as assigned from the source requirements list and are not renumbered by grouping.

B.1 User Account Management

Field	FR-01: User Registration
Description	The system shall allow a new user to register for an account.
Input	Email address, password.
Processing/Behaviour	The system shall validate the email format (see FR-04) and password strength (see FR-07), then create a new user account and associated persistent data record.
Output/Expected Result	A new user account is created and the user is able to log in with the registered credentials.
Preconditions	The email address must not already be registered.
Postconditions	A new user record exists in persistent storage, associated uniquely with that user (see FR-27).

Field	FR-02: User Login
Description	The system shall allow a registered user to log in using their credentials.
Input	Email address, password.
Processing/Behaviour	The system shall authenticate the submitted credentials against stored (hashed) credentials.
Output/Expected Result	On success, the user is granted an authenticated session and access to their task data. On failure, access is denied.
Preconditions	The user must have a registered account.
Postconditions	An authenticated session is established for the user.

Field	FR-03: User Logout
Description	The system shall allow an authenticated user to log out.
Input	Logout action triggered by the user.
Processing/Behaviour	The system shall terminate the user's active session.
Output/Expected Result	The user is signed out and can no longer access protected features until logging in again.
Preconditions	The user must have an active authenticated session.
Postconditions	The session is terminated.

Field	FR-04: Email Format Validation
Description	The system shall validate that email addresses entered during registration (and any other email-entry point) conform to a valid email format.
Input	Email address string.
Processing/Behaviour	The system shall check the input against standard email format rules.
Output/Expected Result	Valid emails are accepted; invalid emails are rejected with an appropriate error indication.
Preconditions	An email address has been submitted.
Postconditions	Only correctly formatted email addresses are accepted for account-related

	operations.

Field	FR-05: View Profile
Description	The system shall allow an authenticated user to view their own profile information.
Input	Request to view profile (authenticated session).
Processing/Behaviour	The system shall retrieve and display the user's profile data.
Output/Expected Result	The user's profile information is displayed.
Preconditions	The user must be authenticated.
Postconditions	None (read-only operation).

Field	FR-06: Edit Profile
Description	The system shall allow an authenticated user to edit their own profile information.
Input	Updated profile field values.
Processing/Behaviour	The system shall validate and persist the updated profile information.
Output/Expected Result	The user's profile is updated and reflects the new values on subsequent views.
Preconditions	The user must be authenticated.
Postconditions	The user's profile record in persistent storage is updated.

Field	FR-07: Password Strength Enforcement
Description	The system shall enforce password-strength requirements when a user sets or changes a password.
Input	Candidate password string.
Processing/Behaviour	The system shall check the candidate password against defined strength rules.

Output/Expected Result	Passwords meeting the requirements are accepted; non-compliant passwords are rejected with an appropriate error indication.
Preconditions	A password is being set, changed, or reset.
Postconditions	Only passwords meeting the strength requirement are stored.
Password Strength Rule	The password shall be a minimum of 8 characters and shall contain at least one uppercase letter, one lowercase letter, one number, and one special character.

Field	FR-08: Change Password
Description	The system shall allow an authenticated user to change their password.
Input	Current password, new password.
Processing/Behaviour	The system shall verify the current password, validate the new password's strength (FR-07), and update the stored (hashed) password.
Output/Expected Result	The user's password is updated; subsequent logins require the new password.
Preconditions	The user must be authenticated and must correctly supply the current password.
Postconditions	The stored password hash is updated.

Field	FR-09: Password Reset
Description	The system shall allow a user who has forgotten their password to reset it via email verification.
Input	Registered email address; verification confirmation; new password.
Processing/Behaviour	The system shall send a verification message to the registered email, confirm the user's identity through that verification, and allow the user to set a new password (subject to FR-07).
Output/Expected Result	The user's password is reset and the user can log in with the new password.
Preconditions	The email address must correspond to a registered account.
Postconditions	The stored password hash is updated.

B.2 Task Management

Field	FR-10: Create Task
Description	The system shall allow an authenticated user to create a new task.
Input	Task title (mandatory), description (optional), due date, priority, category.
Processing/Behaviour	The system shall validate the mandatory title field and store the new task associated with the creating user.
Output/Expected Result	A new task is created and appears in the user's task list.

Preconditions	The user must be authenticated; the task title must be provided.
Postconditions	A new task record exists in persistent storage, associated with the user, with a creation timestamp recorded (see NFR-09).

Field	FR-11: Mark Task as Completed
Description	The system shall allow a user to mark one of their tasks as completed.
Input	Selected task, completion action.
Processing/Behaviour	The system shall update the task's status to completed.
Output/Expected Result	The task is displayed as completed and reflected in dashboard statistics.
Preconditions	The task must exist and belong to the requesting user.
Postconditions	The task's status is updated to completed, with a completion timestamp recorded (see NFR-09).

Field	FR-12: Set Task Due Date
Description	The system shall allow a user to assign a due date to a task.
Input	Selected task, due date value.
Processing/Behaviour	The system shall store the specified due date with the task.
Output/Expected Result	The task's due date is recorded and used for reminders (FR-23), filtering (FR-21), and sorting (FR-22).
Preconditions	The task must exist and belong to the requesting user.

Postconditions	The task's due date attribute is updated.

Field	FR-13: View Task
Description	The system shall allow a user to view the details of their own task.
Input	Selected task.
Processing/Behaviour	The system shall retrieve and display the task's details.
Output/Expected Result	The task's title, description, due date, priority, category, and status are displayed.
Preconditions	The task must exist and belong to the requesting user.
Postconditions	None (read-only operation).

Field	FR-14: Delete Task
Description	The system shall allow a user to permanently delete one of their tasks.
Input	Selected task, delete confirmation.

Processing/Behaviour	The system shall remove the task record from persistent storage.
Output/Expected Result	The task no longer appears in the user's task list.
Preconditions	The task must exist and belong to the requesting user.
Postconditions	The task record is permanently removed from the user's task list, and the deletion action is recorded with a timestamp for traceability.

Field	FR-15: Set Task Description
Description	The system shall allow a user to set an optional description for a task.
Input	Selected task, description text.
Processing/Behaviour	The system shall store the description with the task.
Output/Expected Result	The description is displayed when the task is viewed.

Preconditions	The task must exist and belong to the requesting user.
Postconditions	The task's description attribute is updated.

Field	FR-16: Set Task Title
Description	The system shall require and allow a user to set a mandatory title for a task.
Input	Selected task, title text.
Processing/Behaviour	The system shall validate that the title is non-empty and store it with the task.
Output/Expected Result	The task's title is displayed wherever the task is listed or viewed.
Preconditions	The task must exist and belong to the requesting user (for edits); a title must be supplied for creation.
Postconditions	The task's title attribute is set or updated.

Field	FR-17: Set Task Priority
Description	The system shall allow a user to assign a priority level to a task.
Input	Selected task, priority value.
Processing/Behaviour	The system shall store the specified priority with the task.
Output/Expected Result	The task's priority is displayed and used for filtering (FR-21) and sorting (FR-22).
Preconditions	The task must exist and belong to the requesting user.
Postconditions	The task's priority attribute is set.
Allowed Values	Low, Medium, High.

Field	FR-18: Update Task Priority
Description	The system shall allow a user to update the priority of an existing task.
Input	Selected task, new priority value.
Processing/Behaviour	The system shall replace the task's existing priority value with the new one.
Output/Expected	The task reflects the updated priority in subsequent views, filters, and sorts.

Result	
Preconditions	The task must exist and belong to the requesting user.
Postconditions	The task's priority attribute is updated.

Field	FR-19: Assign Task Category
Description	The system shall allow a user to assign a category to a task.
Input	Selected task, selected category (from user's existing categories, see FR-20).
Processing/Behaviour	The system shall associate the selected category with the task.
Output/Expected Result	The task's category is displayed and used for filtering (FR-21).
Preconditions	The task must exist and belong to the requesting user; the category must exist for that user.
Postconditions	The task's category attribute is set.

Field	FR-20: Manage Categories
Description	The system shall allow a user to create, rename, and delete their own task categories.
Input	Category name (for creation/renaming), selected category (for renaming/deletion).
Processing/Behaviour	The system shall create a new category, update an existing category's name, or remove a category, as requested.
Output/Expected Result	The user's list of available categories reflects the requested change.
Preconditions	The user must be authenticated; for renaming/deletion, the category must belong to the requesting user.
Postconditions	The category record is created, updated, or removed in persistent storage. When a category is deleted, the system shall remove the category assignment from all tasks that referenced it, leaving those tasks uncategorized; the tasks themselves shall not be deleted.

Field	FR-21: Filter Tasks

Description	The system shall allow a user to filter their task list based on task attributes.
Input	Selected filter criteria (status, priority, category, due date).
Processing/Behaviour	The system shall apply the selected filter criteria to the user's task list.

Output/Expected Result	Only tasks matching the selected criteria are displayed.
Preconditions	The user must be authenticated.
Postconditions	None (the underlying task data is unchanged; only the displayed view is affected).

Field	FR-22: Sort Tasks
Description	The system shall allow a user to sort their task list based on task attributes.
Input	Selected sort attribute (status, priority, category, or due date) and sort order.
Processing/Behaviour	The system shall reorder the displayed task list according to the selected attribute and order.
Output/Expected Result	The task list is displayed in the requested order.
Preconditions	The user must be authenticated.
Postconditions	None (the underlying task data is unchanged; only the displayed view is affected).

Field	FR-23: Task Due Date Reminder
Description	The system shall notify a user when a task's due date is approaching.
Input	Task due date, current date/time.
Processing/Behaviour	The system shall monitor task due dates and generate a reminder as the due date approaches.
Output/Expected Result	The user receives a reminder for the relevant task.
Preconditions	The task must have a due date set (FR-12).
Postconditions	None.

Reminder Lead Time	The system shall generate a reminder 24 hours before a task's due date.
Delivery Mechanism	The reminder shall be delivered as a mobile push notification.

B.3 Dashboard and Progress Tracking

Field	FR-24: Dashboard Statistics Overview
Description	The system shall present summary statistics of the user's tasks on a dashboard.
Input	Request to view dashboard (authenticated session).
Processing/Behaviour	The system shall compute summary statistics (e.g., counts of tasks by status) from the user's task data.
Output/Expected Result	Summary statistics are displayed on the dashboard within the performance threshold defined in NFR-01.
Preconditions	The user must be authenticated.

Postconditions	None (read-only operation).

Field	FR-25: Dashboard Graphical Overview
Description	The system shall present a graphical representation of the user's task progress on the dashboard.
Input	Request to view dashboard (authenticated session).
Processing/Behaviour	The system shall generate graphical representations (e.g., charts) of the user's task progress data.
Output/Expected Result	Graphical progress information is displayed on the dashboard within the performance threshold defined in NFR-01.
Preconditions	The user must be authenticated.
Postconditions	None (read-only operation).
Graph Types	The dashboard shall present a pie chart showing task status distribution and a bar chart showing task counts by priority (Low, Medium, High).

B.4 Data Persistence and Session Management

Field	FR-26: Persistent Data Storage
Description	The system shall persistently store user and task data so that it remains available across sessions.
Input	User and task data generated through normal system use.
Processing/Behaviour	The system shall write and retain data in persistent storage.
Output/Expected Result	Data created in one session is available in subsequent sessions.
Preconditions	None.
Postconditions	Data remains available after the session ends.

Field	FR-27: Data Association Per User
Description	The system shall associate all task and profile data with the specific user that owns it.
Input	User identity (from authenticated session), data being created or accessed.
Processing/Behaviour	The system shall tag stored data with the owning user's identity and restrict retrieval/modification accordingly.
Output/Expected Result	Each user's data is distinguishable from other users' data.
Preconditions	The user must be authenticated.
Postconditions	Data records remain correctly associated with their owning user.

Field	FR-28: Session Timeout
Description	The system shall automatically terminate a user's session after a defined period of inactivity.
Input	Session activity/inactivity timing.
Processing/Behaviour	The system shall monitor session activity and terminate the session once the inactivity threshold is reached.
Output/Expected Result	The user is logged out automatically and must re-authenticate to regain access.
Preconditions	The user must have an active authenticated session.

Postconditions	The session is terminated.
Inactivity Threshold	The system shall terminate a session after 30 minutes of user inactivity.

Non-Functional Requirements

Field	NFR-01: Dashboard Loading Performance
Description	Performance requirement for dashboard responsiveness.
Requirement	The system shall load the dashboard (FR-24, FR-25) within three seconds under normal operating conditions.
Verification/Measurement	Measured by timing dashboard load under normal load conditions across repeated test runs.

Field	NFR-02: Hashed Password Storage
Description	Security requirement for password storage.
Requirement	The system shall store all user passwords using one-way hashing and shall not store passwords in plaintext.
Verification/Measurement	Verified by inspecting the stored password data to confirm it is hashed and non reversible.

Field	NFR-03: Authenticated Access
Description	Security requirement restricting task-related features to authenticated users.
Requirement	The system shall require successful authentication before granting access to task related features.
Verification/Measurement	Verified by attempting to access task features without a valid authenticated session and confirming access is denied.

Field	NFR-04: Prevention of Cross-User Data Access
Description	Data integrity/authorization requirement preventing users from accessing other users' data.

Requirement	The system shall prevent a user from viewing or modifying another user's task or profile data.
Verification/Measurement	Verified through test cases attempting cross-user data access and confirming such attempts are rejected.

Field	NFR-05: No Data Loss
Description	Data integrity requirement preserving submitted task operations.
Requirement	The system shall preserve all successfully submitted task operations without data loss.
Verification/Measurement	Verified through test cases confirming that submitted create/update/delete operations are correctly and durably reflected in storage.

Field	NFR-06: Data Unaltered Except by User Action
Description	Data integrity requirement preventing unauthorized or unintended data changes.
Requirement	The system shall ensure that stored data is not altered except as a result of an explicit action by the owning user.
Verification/Measurement	Verified through test cases confirming that data remains unchanged in the absence of a corresponding user action.

Field	NFR-07: Backup and Recovery
Description	Reliability requirement for data backup and restoration.
Requirement	The system shall support manual database backup and restoration.
Verification/Measurement	Verified by performing a manual backup, simulating data loss, and confirming successful restoration from the backup.

Field	NFR-08: Support for High Task Volume (Scalability)
Description	Scalability requirement for per-user task volume.
Requirement	The system shall support at least 1,000 tasks per user without exceeding the dashboard performance threshold defined in NFR-01.

Verification/Measurement	Verified by populating a test account with 1,000 tasks and measuring dashboard load time against the NFR-01 threshold.

Field	NFR-09: Timestamps for User Actions (Traceability)
Description	Traceability requirement for recording key task actions.
Requirement	The system shall record a timestamp for key task actions, including creation, editing, completion, and deletion.
Verification/Measurement	Verified by performing each action and confirming a corresponding timestamp is recorded and retrievable.

Design Constraints

D.1 Security Constraints

• Passwords shall be stored using one-way hashing; plaintext password storage is not permitted (NFR-02). • Access to task-related features shall require authentication (NFR-03).

• User sessions shall be automatically terminated after 30 minutes of inactivity (FR-28).

D.2 Database/Data Integrity Constraints

• User and task data shall be persistently stored and shall remain available across sessions (FR-26). • Each data record shall be associated with a single owning user, and access shall be restricted accordingly (FR 27, NFR-04).

• Submitted task operations shall not result in data loss (NFR-05), and data shall not be altered except through explicit user action (NFR-06).

• Key task actions shall be timestamped for traceability (NFR-09).

D.3 Performance/Resource Constraints

• The dashboard shall load within three seconds under normal conditions (NFR-01).

• The system shall support at least 1,000 tasks per user without exceeding the dashboard performance threshold (NFR-08).

D.4 Operating Environment Constraints

The system shall be implemented and delivered as an Android mobile application and shall operate on compatible Android devices.

D.5 Other Implementation Constraints

• The system shall support manual database backup and restoration.